University of Applied Sciences

HOCHSCHULE EMDEN·LEER

Algorithms

Prof. Dr. E. Wings

Agenda

1. Roots of a Function
2. Conclusion

Roots of a Function

Roots of a Function

A standard problem is finding the roots of a function f:

$$f(x) = 0$$

Analytical Solution

Normally, you prefer an analytical solution.

Analytical Solution

A cubic equation has the form

$$a \cdot x^3 + b \cdot x^2 + c \cdot x + d = 0, \qquad a \neq 0$$

The general solution is:

$$\text{with} \quad \Delta_0 = b^2 - 3ac, \quad \Delta_1 = 2b^3 - 9abc + 27a^2 d,$$

$$C = \sqrt[3]{\frac{\Delta_1 \pm \sqrt{\Delta_1^2 - 4\Delta_0^3}}{2}}, \quad \zeta = \frac{-1 + \sqrt{-3}}{2}$$

we get:

$$x_k = -\frac{1}{3a}\left(b + \zeta^k C + \frac{\Delta_0}{\zeta^k C}\right), \quad k = 0, 1, 2$$

Analytical Solution

Problems:

- One, two or three solutions?
- Simple equation, but complex solutions
- Complex equation, perhaps there is no analytical soltion
- How to handle complex numbers?
- Accuracy?
- How many solutions?

Newton Algorithm

A standard algorithm is the Newton algorithm. Find the root of the tangent t at a position x_0 instead of the function:

$$t(x) = f'(x_0)(x - x_0) + f(x_0) = 0$$

$$\Leftrightarrow \quad x = x_0 - \frac{f(x_0)}{f'(x_0)}$$

Newton algorithm:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Newton Algorithm - Advantages

- A simple algorihtm
- A fast algorithm (normally)

Disadvantages

- How do I get the derivative?
- Can I derivate a function automatically?
- Can I approximate a derivative?
- What happens near a root of the derivative?
- What happen if there is no root, e.g. $x^2 + 1$?
- Which root will you find?
- How do I choose the start value?
- What happens when the algorithm oscillates?
- When does the algorithm stop?

How I get the derivative?

- The calculation of a derivative is not always easy.
- I have to program the derivative, too.

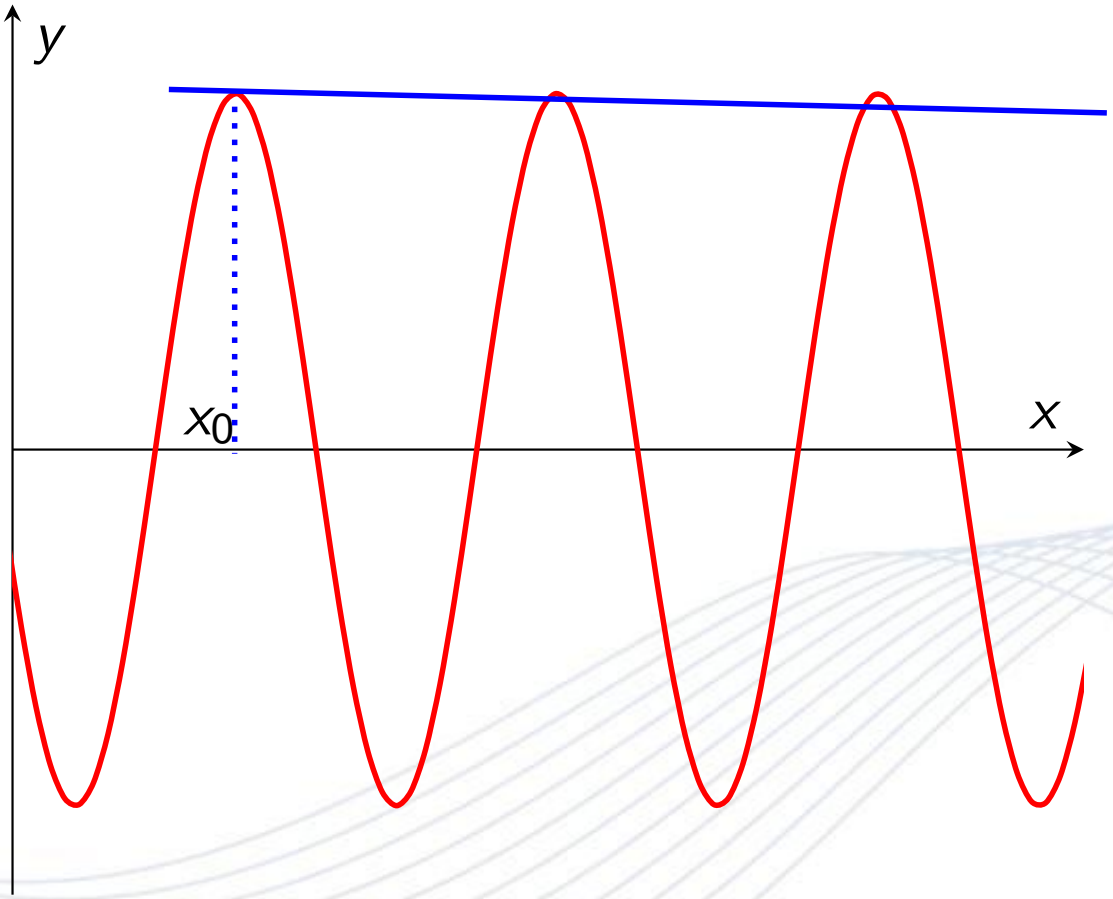
Can I derivate a function automatically?

- There exist some nice tools: Maple, MatLab
- It's not possible to program by yourself for all functions
  $\Rightarrow$ Restrictions

Can I approximate a derivative?

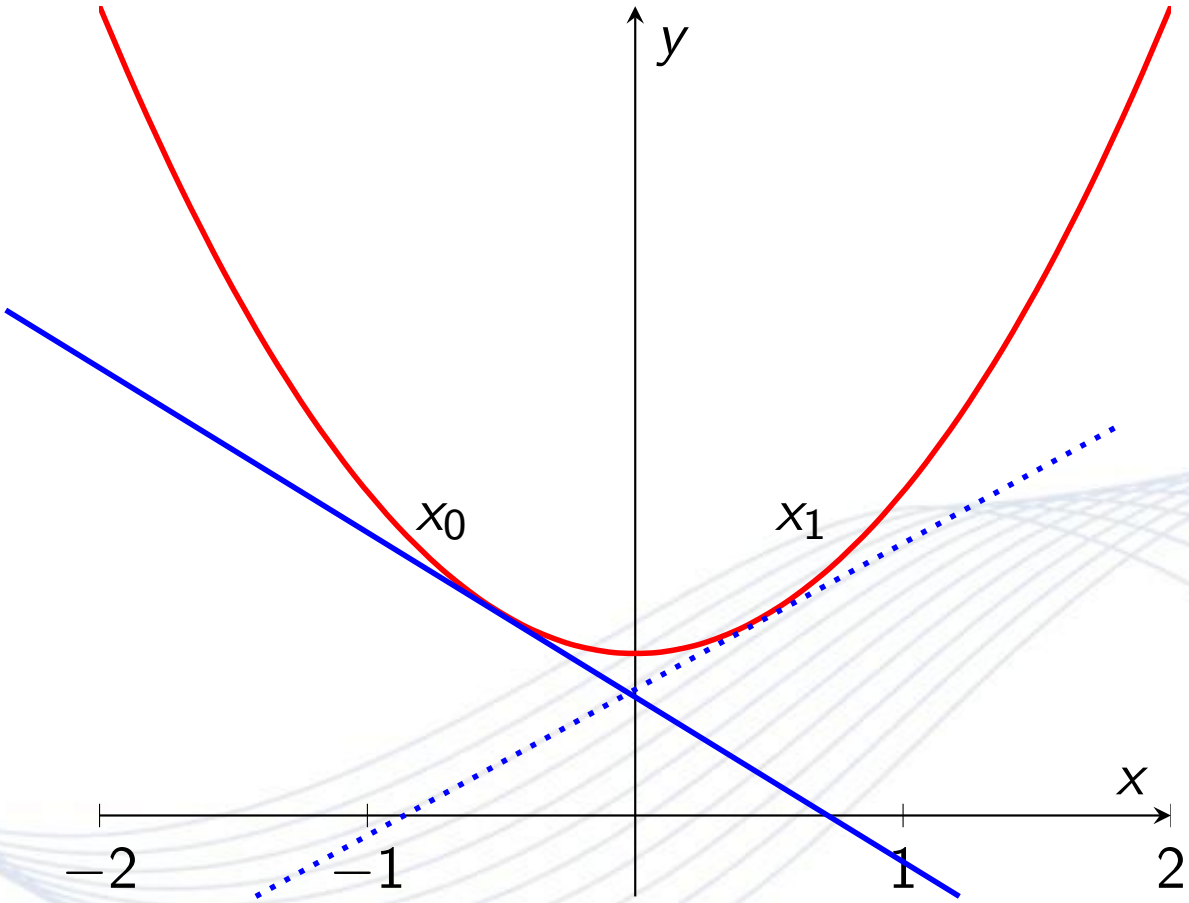
- New algorithm: Bisection Method
- It's slower.
- It needs an additional start value.

What happens near a root of the derivative?

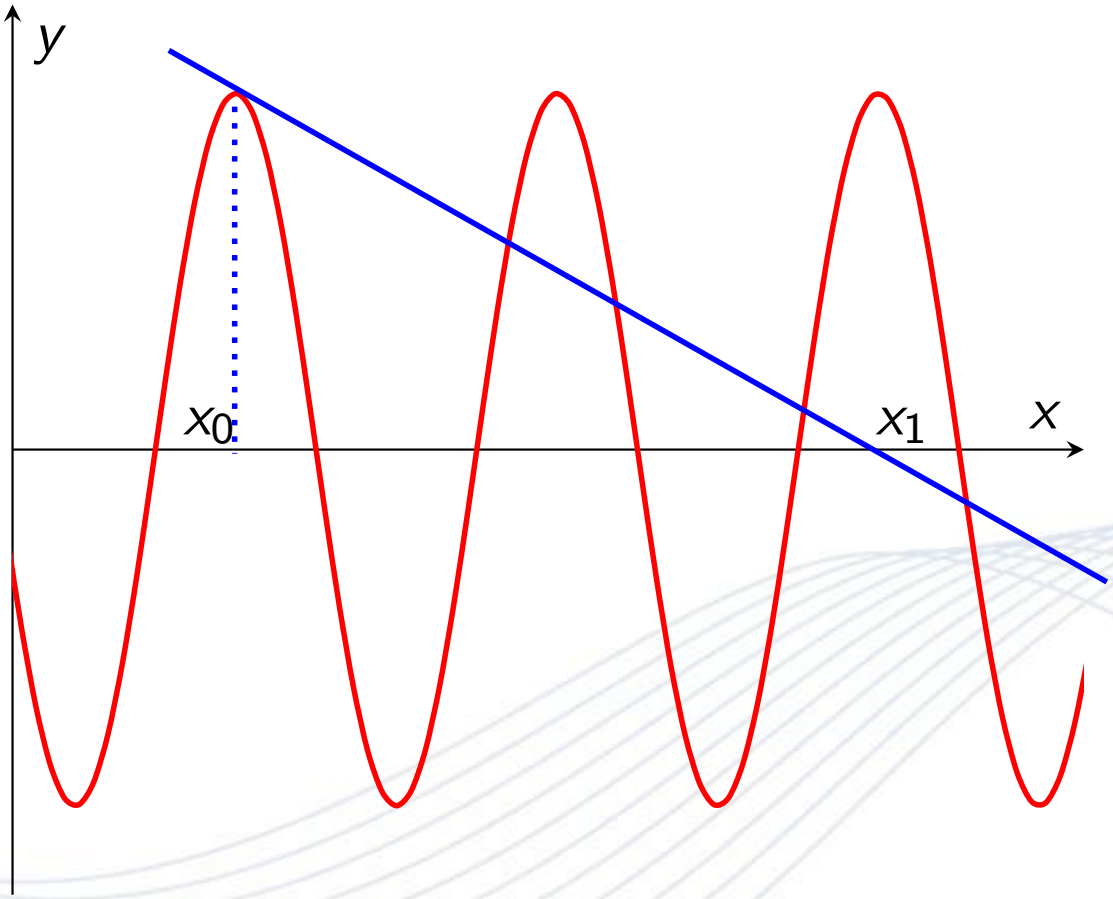


What happen if there is no root?

Example: $f(x) = x^2 + 1$



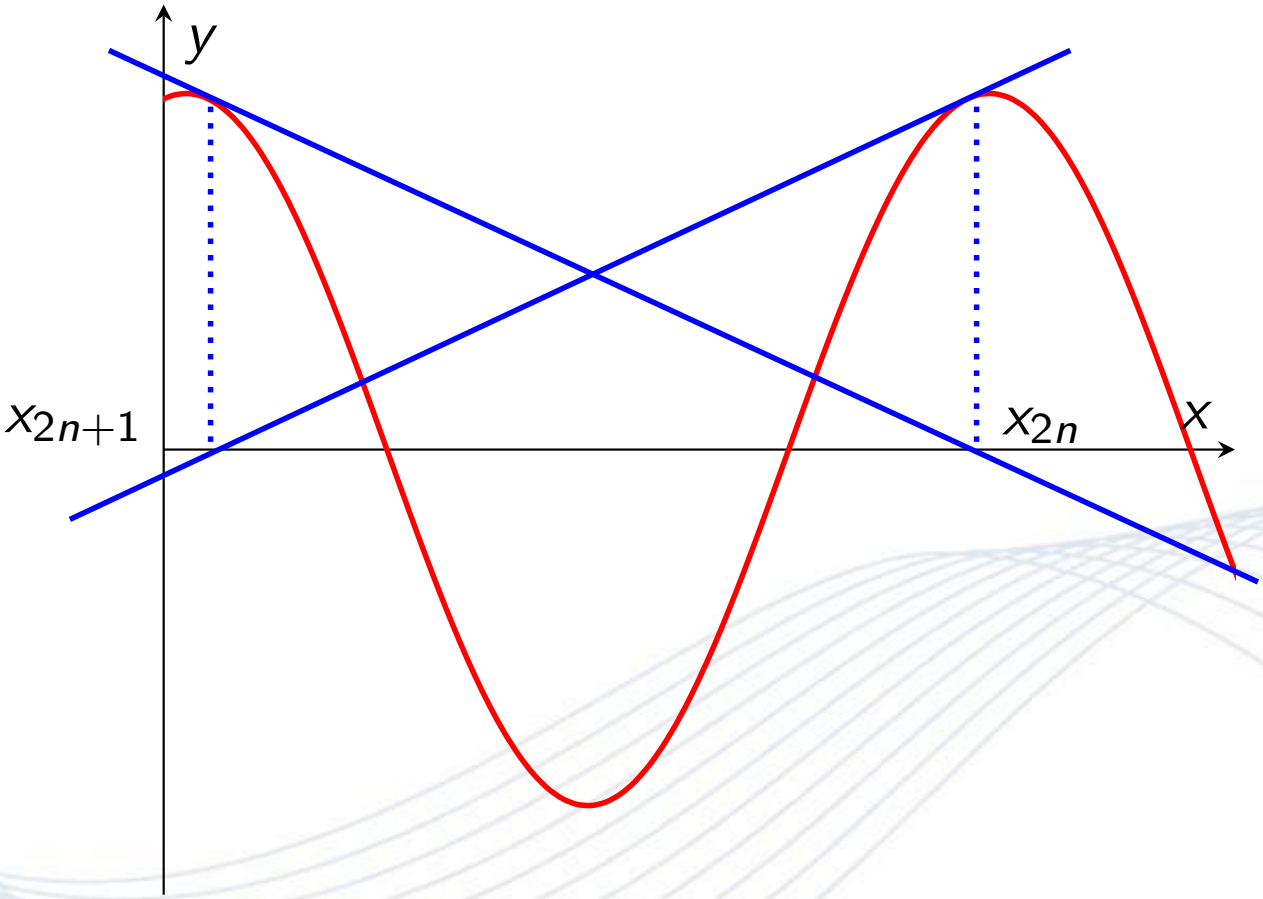
Which root will you find?



How do I choose the start value?

- If you choose a wrong start value, the algorithm is not convergent.
- Simple rule: If the start value is more nearer to the root, the start value is better.
- Simple rule: Try some values and choose the best one.

What happens when the algorithm oscillates?



When does the algorithm stop?

- You have to define some rules to stop the algorithm. Standard rules are:

  1. Maximal steps for n
  2. $$|x_n - x_{n+1}| < \delta$$
  3. $$|f(x_n) - f(x_{n+1})| < \varepsilon_1$$
  4. $$|f(x_n)| < \varepsilon_2$$

General Algorithm

The Newton Method is a simple, well known algorithm.

What happens if you want to solve more complex problems with sophisticated algorithms?

Conclusion

Conclusion

There are different cases:

- Standard solution
- Division by zero
- No solution
- Root jumping
- Oszillation

In General: Identify the different Cases

- Find the criteria for the standard case.
- Find the cases with problems/specialities

$\Rightarrow$ Make a list

In General: Identify the different Cases

- Design test cases for the standard case.
- Design for each case with problmes special tests
- Design tests at random

$\Rightarrow$ Make a list

Stop Criteria

- Design a simple stop criteria, e.g. maximal number of steps
- Design a stop criteria for a solution
- Desgin stop criteria for special cases